

Trade Behavior Analysis Under Market Sentiment

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Introduction

This report analyzes how trader behavior and performance vary across different market sentiment regimes using historical trade data and bitcoin fear and greed index that was provided to me. The goal is to uncover behavioral patterns that can inform smarter trading strategies.

Data Description

Historical trader data consisted of execution price, trade size, direction, PnL, and fees

Market sentiment into fear, greed, neutral, extreme fear, and extreme greed

Sentiment categories were simplified into fear/greed/neutral for clarity

Methodology

- Data cleaning and timestamp normalization
 - Merging datasets on trade date
 - Use of trade size (USD) and execution behavior as proxies for risk exposure
- (leverage was mentioned in documentation but not present in the actual dataset)

Key Findings

Profitability and Volatility

- Trading during Fear-dominated periods exhibits significantly higher volatility, with both the largest profits and the largest losses observed in the dataset.
- Although the average PnL during Fear is marginally higher than Greed, the risk associated with these trades is substantially greater.
- Greed periods demonstrate lower PnL volatility, with fewer extreme outcomes.
- Median PnL across all sentiment regimes remains close to zero, suggesting that most trades are small or near break-even.

Risk Exposure (Trade Size as Proxy)

- Average trade size is highest during Fear, indicating increased risk-taking during fearful market conditions.
- Fear periods also contain extreme trade-size outliers, suggesting panic-driven or forced execution behavior.
- Trades executed during Greed and Neutral periods are comparatively smaller and more controlled.

Directional Trading Behavior

- During Fear, traders predominantly engage in opening new positions, particularly long positions.
- Greed periods show a more balanced mix of position openings and closures, including a higher proportion of short-position closures, consistent with profit-taking behavior.
- Neutral periods exhibit lower overall activity.

Trading Costs

- Average transaction fees are highest during Fear, reflecting increased trading urgency and aggressive execution.
- Fees decline during Greed and are lowest during Neutral sentiment periods.

Overall Insight

- Market sentiment strongly influences trader behavior.
- Fear amplifies both risk-taking and outcome dispersion, while Greed corresponds to more controlled trading behavior with reduced extremes.
- These findings highlight the importance of incorporating sentiment-aware risk management into trading strategies.

Conclusion

The analysis indicates that market sentiment strongly influences trader behavior. Fear-driven markets exhibit higher risk-taking, volatility, and aggressive execution, while Greed periods demonstrate relatively controlled behavior and profit realization. These insights highlight the importance of sentiment-aware risk management strategies

